



PRODUCT DESCRIPTION:
White to off white powder filled in size 2 hard gelatin capsule with cream cap & brown body printed OP on cap and 75 on body in black.

COMPOSITION:
Each capsule contains Osetlamivir phosphate equivalent to Osetlamivir 75mg.

PHARMACOLOGY:
Osetlamivir is an ethyl ester prodrug requiring ester hydrolysis for conversion to the active form, osetlamivir carboxylate. The proposed mechanism of action of osetlamivir is inhibition of influenza virus neuraminidase with the possibility of alteration of virus particle aggregation and release.
The concentrations of osetlamivir carboxylate required for inhibition of influenza virus were highly variable depending on the assay method used and the virus tested. The 50% and 90% inhibitory concentrations (IC 50 and IC 90) were in the range of 0.0008 µM to >35µM and 0.004µM to >100 µM, respectively (1 µM=0.284 µg/mL). The relationship between the in vitro antiviral activity in cell culture and the inhibition of influenza virus replication in humans has not been established.

PHARMACOKINETICS:
Absorption: Osetlamivir is readily absorbed from the GIT after oral administration of osetlamivir phosphate (pro-drug) and is extensively converted by predominantly hepatic esterases to the metabolite (osetlamivir carboxylate). At least 75% of an oral dose reaches the systemic circulation as the metabolite. Exposure to the pro-drug is <5% relative to the active metabolite. Plasma concentrations of both pro-drug and active metabolite are proportional to dose and are unaffected by co-administration with food. *Distribution:* The mean volume of distribution (Vss) of the active metabolite is approximately 23L in humans, a volume equivalent extracellular body fluid. Since neuraminidase activity is extracellular, osetlamivir carboxylate distributes to all sites of influenza virus spread. The binding of the osetlamivir carboxylate to human plasma protein is negligible (approximately 3%). *Metabolism:* Osetlamivir is extensively converted to osetlamivir carboxylate by esterases located predominantly in the liver. *In vitro* studies demonstrated that neither osetlamivir nor the active metabolite is a substrate for, or an inhibitor of, the major cytochrome P450 isoforms. No phase 2 conjugates of either compound have been identified *in vivo*. *Elimination:* Absorbed osetlamivir is primarily (>90%) eliminated by conversion to osetlamivir carboxylate. It is not further metabolised and is eliminated in the urine. Peak plasma concentrations of osetlamivir carboxylate decline with a half-life of 6-10 hours in most subjects. The active metabolite is eliminated entirely by renal excretion. Renal clearance (18.81/h) exceeds glomerular filtration rate (7.51/h) indicating that tubular secretion occurs in addition to glomerular filtration. Less than 20% of an oral radiolabelled dose is eliminated in faeces.

INDICATIONS:
Treatment of Influenza
Osetlamivir Capsules is indicated for the treatment of influenza in adults and children ≥ 1 year of age. Treatment should commence as soon as possible but no later than 48 hours after the onset of the initial symptoms of infection.
Prophylaxis of Influenza
Osetlamivir Capsules is indicated for post exposure prophylaxis of influenza for

adults and children ≥ 1 year following contact with a clinically diagnosed influenza case when influenza virus is circulating in the community.

DOSEAGE:
Osetlamivir Capsules may be taken with or without food. However, when taken with food, tolerability may be enhanced in some patients.

Standard Dosage - Treatment of Influenza:
Treatment should begin within the 1st or 2nd day of onset of symptoms of influenza.
Adults and Adolescents: The recommended oral dose for treatment of influenza in adults and adolescents ≥ 13 years is 75 mg twice daily for 5 days.
Children: Children weighing > 40 kg who are able to swallow capsules may also receive treatment with a 75mg capsule twice daily.

Standard Dosage - Prophylaxis of Influenza: The recommended oral dose of Osetlamivir for prophylaxis of influenza in adults and adolescents ≥ 13 years following close contact with an infected individual is 75 mg once daily for 10 days. Therapy should begin within 2 days of exposure. The recommended dose for prophylaxis during a community outbreak of influenza is 75 mg once daily. Safety and efficacy have been demonstrated for up to 6 weeks. The duration of protection lasts for as long as dosing is continued. **Children:** Children weighing > 40 kg, who are able to swallow capsules, may also receive prophylaxis with a 75 mg capsule once daily.

Special Dosage Instructions
Hepatic Impairment: No dose adjustment is required for patients with mild or moderate hepatic dysfunction in the treatment or prophylaxis of influenza. The safety and pharmacokinetics in patients with severe hepatic impairment have not been studied.
Renal Impairment:
Treatment of influenza:No dose adjustment is necessary for patients with creatinine clearance above 30 mL/min. In patients with a creatinine clearance of 10-30 mL/min, it is recommended that the dose be reduced to 75 mg once daily for 5 days. No dosing recommendation is available for patients undergoing routine haemodialysis and continuous peritoneal dialysis with end stage renal disease and for patients with creatinine clearance ≤10mL/min.
Prophylaxis of influenza: No dose adjustment is necessary for patients with creatinine clearance >30 mL/min. In patients with creatinine clearance of 10-30 mL/min, it is recommended that the dose be reduced to 75mg of capsule every other day. No dosing recommendation is available for patients undergoing routine haemodialysis and continuous peritoneal dialysis with end stage renal disease and for patients with creatinine clearance ≤10mL/min.
Geriatric Patient: No dose adjustment is required for elderly patients in the treatment or prophylaxis of influenza.
Children: The safety and efficacy of osetlamivir in children under 1 year has not been established.

Clinical Trials
Adult Treatment Studies: In phase III clinical studies in the treatment of influenza, the most frequently reported adverse events were nausea and vomiting. These events were transient and generally occurred with first dosing. These events did not lead to patient discontinuation of study drug in the vast majority of instances. Those events with an incidence of ≥ 1% and which were reported more frequently in patients taking osetamivir compared with placebo, irrespective of causality, were



Size: 160 x 150 mm

Front Side

nausea, vomiting, abdominal pain and headache. Other events reported are diarrhoea, bronchitis, dizziness and insomnia.

Adult Prophylaxis studies: Adverse events were qualitatively very similar to those seen in the treatment studies, despite a longer duration of dosing. Events reported more frequently in subjects receiving oseltamivir compared to subjects receiving placebo in prophylaxis studies, and more commonly than in treatment studies, were aches and pains, rhinorrhea, dyspepsia and upper respiratory tract infections. However, the difference in incidence between oseltamivir and placebo for these events was less than 1%. There were no clinically relevant differences in the safety profile of the elderly subjects, who received oseltamivir or placebo, compared with the younger population.

Paediatric treatment studies: Adverse events occurring in > 1% of children receiving oseltamivir are vomiting, diarrhoea, otitis media, abdominal pain, asthma, nausea, epistaxis, pneumonia, ear disorder, sinusitis, bronchitis, conjunctivitis, dermatitis, lymphadenopathy and tympanic membrane disorder. The most frequently reported adverse event was vomiting. These events generally occurred once, resolved despite continued dosing and did not cause discontinuation of treatment in the vast majority of cases.

Paediatric Prophylaxis: Gastrointestinal events were the most frequent, particularly vomiting.

Post-Marketing

Skin and subcutaneous tissue disorder: rare cases of hypersensitivity reactions such as allergic skin reactions including dermatitis, rash, eczema, urticaria, and very rare cases of erythema multiforme and Stevens-Johnson-Syndrome and toxic epidermal necrolysis are reported. Also, allergy, anaphylactic/anaphylactoid reactions and face edema are reported rarely. **Liver and biliary system disorder:** very rare reports of hepatitis and elevated liver enzymes have been reported in patients with influenza like illness receiving oseltamivir.

Psychiatric disorders/Nervous system disorders: Convulsion and delirium (including symptoms such as altered level of consciousness, confusion, abnormal behaviour, delusions, hallucinations, agitation, anxiety, nightmares) have been reported during oseltamivir administration in patients with influenza, predominately in children and adolescents. In rare cases, these events resulted in accidental injury. The contribution of oseltamivir to those events is unknown. Such neuropsychiatric events have also been reported in patients with influenza who were not taking oseltamivir.

Gastro-intestinal disorders: In rare cases gastro-intestinal bleedings were observed after the use of oseltamivir. In particular, haemorrhagic colitis was reported that subsided when the course of influenza abated or treatment with oseltamivir was interrupted.

PRECAUTIONS: Convulsion and delirium like neuropsychiatric events have been reported during oseltamivir administration in patients with influenza, predominately in children and adolescent. In rare cases these events resulted in accidental injury. The contribution of oseltamivir to these events are unknown and these have also been reported in patients with influenza who were not taking oseltamivir. Patients especially children and adolescent, should be closely monitored for signs of abnormal behavior. There is no evidence for efficacy of oseltamivir in any illness caused by agents other than influenza viruses Types A and B.

Dose adjustment is recommended for patients with creatinine clearance of 10-30mL/min for the treatment of influenza and prophylaxis of influenza. No dosing recommendation is available for patients undergoing routine haemodialysis and continuous peritoneal dialysis with end stage renal disease and for patients with creatinine clearance ≤10mL/min.

USE IN PREGNANCY AND LACTATION:

Pregnancy: There are insufficient human data upon which to base an evaluation of risk of oseltamivir to the pregnant woman or developing fetus. Because animal reproductive studies may not be predictive of human response and there are no adequate and well-controlled studies in pregnant women, oseltamivir should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Lactation: In lactating rats, oseltamivir and oseltamivir carboxylate are excreted in the milk. It is not known whether oseltamivir or oseltamivir carboxylate is excreted in human milk. Oseltamivir should, therefore, be used only if the potential benefit for the lactating mother justifies the potential risk to the breast-fed infant.

CONTRAINDICATIONS: Oseltamivir Capsules is contraindicated in patients with known hypersensitivity to any of the components of the product.

DRUG INTERACTIONS: Information derived from pharmacology and pharmacokinetic studies of oseltamivir suggests that clinically significant drug interactions are unlikely. Oseltamivir is extensively converted to oseltamivir carboxylate by esterases, located predominantly in the liver. Drug interactions involving competition for esterases have not been extensively reported in literature. Low protein binding of oseltamivir and oseltamivir carboxylate suggests that the probability of drug displacement interactions is low. In vitro studies demonstrate that neither oseltamivir nor oseltamivir carboxylate is a good substrate for P450 mixed-function oxidases or for glucuronyl transferases. Co-administration of probenecid results in an approximate twofold increase in exposure to oseltamivir carboxylate due to a decrease in active anionic tubular secretion in the kidney. However, due to the safety margin of oseltamivir carboxylate, no dose adjustments are required when co-administering with probenecid.

Adverse Effects/Undesirable Effects:

Blood and lymphatic system disorders
Frequency 'Rare' : Thrombocytopenia

OVERDOSE AND TREATMENT: At present, there has been no experience with overdose. Single doses of up to 1000 mg of oseltamivir have been associated with nausea and/or vomiting.

AVAILABILITY: Blister pack of 1 x 10 capsules.

STORAGE Store below 30°C. Keep out of reach of children.

For further information, please consult your physician or pharmacist.

Manufactured by:

Strides Pharma Science Limited

S- No.36/7, Suragajakkanahalli,

Indlavadi Cross, Anekal Taluk, Bangalore, India

Product Registration Holder and Distributor in Malaysia:

GERMAX SDN BHD

No.: 10, Jalan PJU 3/48, Sunway Damansara Technology Park,
47810 Petaling Jaya Selangor, MALAYSIA.

DATE OF REVISION:

17-10-2022



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Size: 160 x 150 mm